General Trees

A graph which has no cycle is called an acyclic graph. A tree is an acyclic graph or graph having no cycles.

A tree or general trees is defined as a non-empty finite set of elements called vertices or nodes having the property that each node can have minimum degree 1 and maximum degree n. It can be partitioned into n+1 disjoint subsets such that the first subset contains the root of the tree and remaining n subsets includes the elements of the n sub tree.

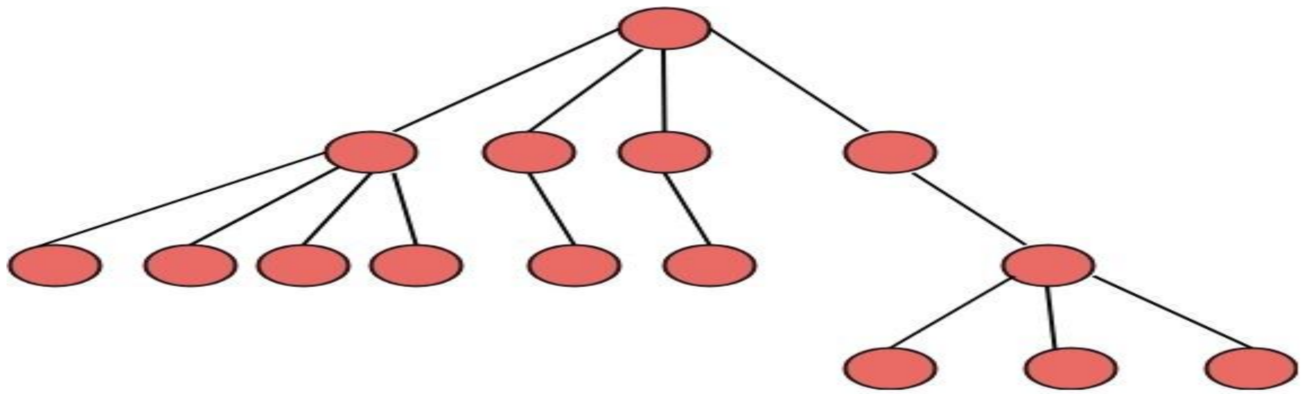
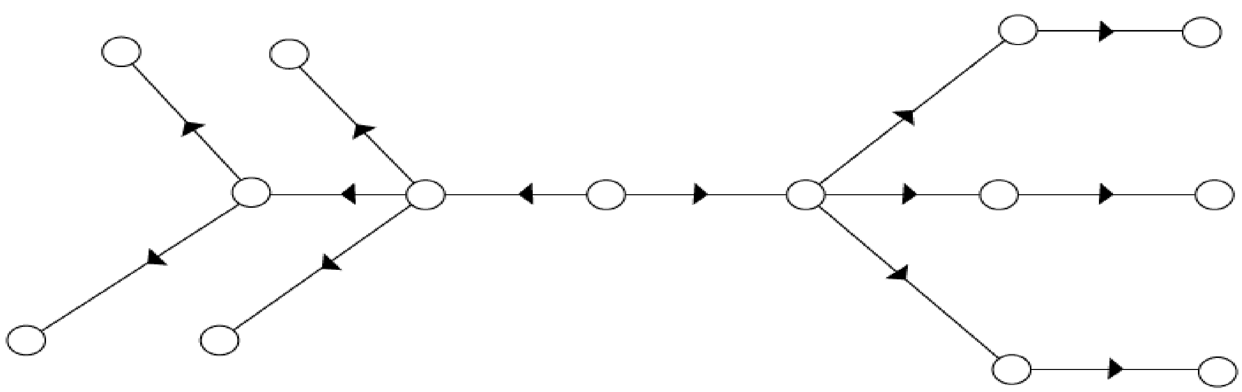


Fig:General Trees

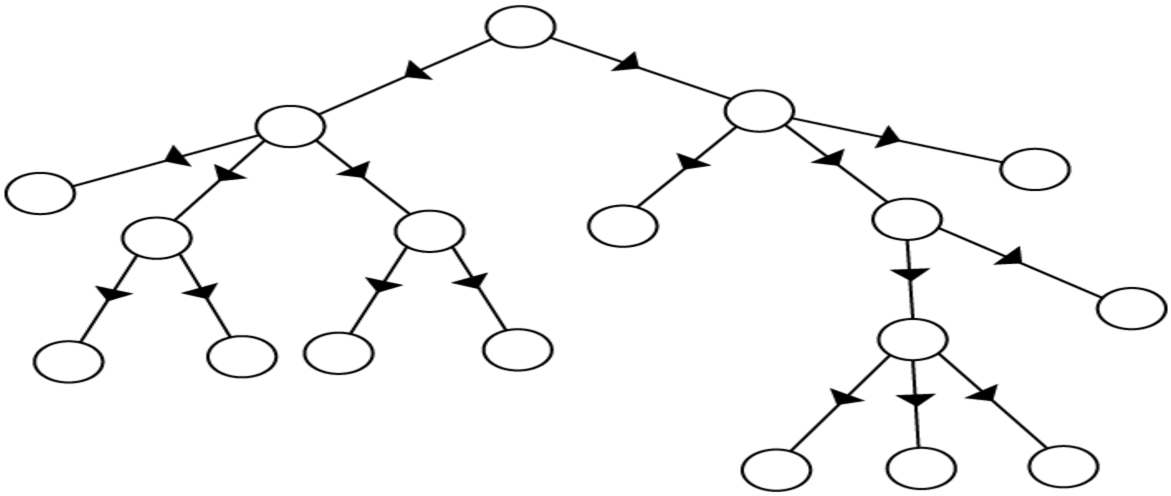
Directed Trees:

A directed tree is an acyclic directed graph. It has one node with in degree 1, while all other nodes have in degree 1 as shown in fig:



Directed Trees

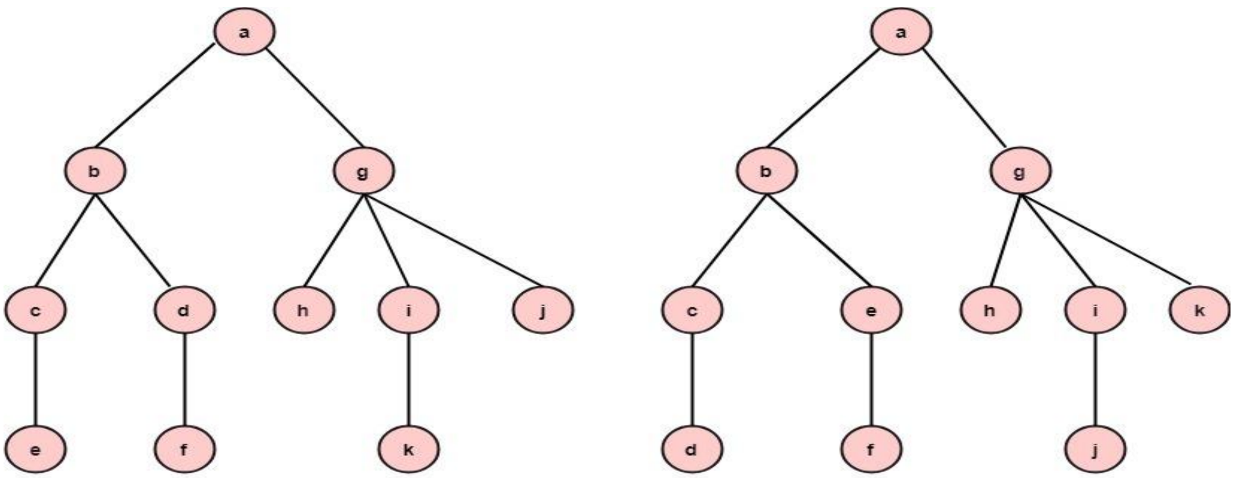
The node which has out degree 0 is called an external node or a terminal node or a leaf. The nodes which have out degree greater than or equal to one are called internal node.



Ordered Trees:

If in a tree at each level, an ordering is defined, then such a tree is called an ordered tree.

Example: The trees shown in the figures represent the same tree but have different orders.



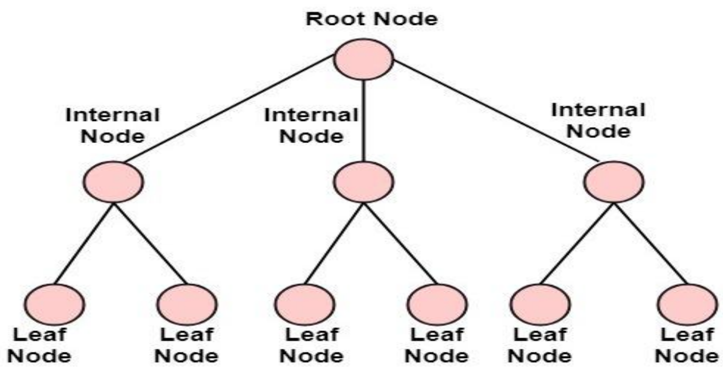
Properties of Trees:

1. There is only one path between each pair of vertices of a tree.
2. If a graph G there is one and only one path between each pair of vertices G is a tree.
3. A tree T with n vertices has n-1 edges.
4. A graph is a tree if and only if it a minimal connected.

Rooted Trees:

If a directed tree has exactly one node or vertex called root whose incoming degrees is 0 and all other vertices have incoming degree one, then the tree is called rooted tree.

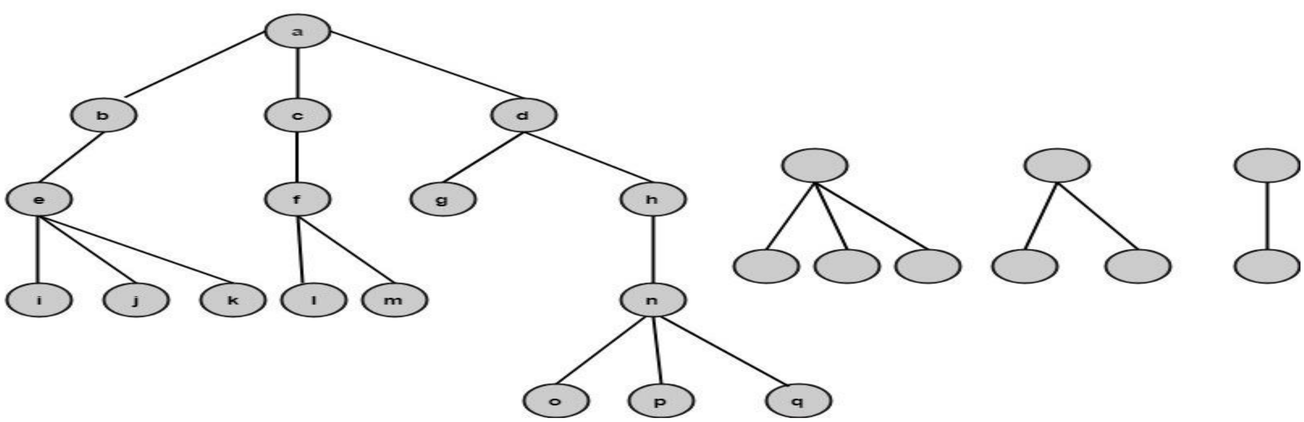
A rooted tree is a connected acyclic graph with a special node that is called the root of the tree and every edge directly or indirectly originates from the root. An ordered rooted tree is a rooted tree where the children of each internal vertex are ordered. If every internal vertex of a rooted tree has not more than m children, it is called an m-ary tree. If every internal vertex of a rooted tree has exactly m children, it is called a full m-ary tree. If m=2, the rooted tree is called a binary tree.



Path length of a Vertex:

The path length of a vertex in a rooted tree is defined to be the number of edges in the path from the root to the vertex.

Example: Find the path lengths of the nodes b, f, l, q as shown in fig:



Solution: The path length of node b is one.
The path length of node f is two.
The path length of node l is three.
The path length of the node q is four.